

[Aim: 100/100 in Maths]

आरंभ

~~Def~~ → Similarity

TRIANGLES

Lecture #5

Given $\rightarrow$
 To prove $\rightarrow$
#LP: In the Figure, PA, QB and RC are perpendicular to AC. Prove that

$$\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$$

Since $\angle PAB = \angle QBC = 90^\circ$

$\therefore PA \parallel QB$

$\angle PAB = \angle QBC$ (both 90°)

$\angle 1 = \angle 2$ (corr. $\angle$ s)

AA
 $\triangle CBQ \sim \triangle CAP$

$$\frac{CB}{CA} = \frac{BQ}{AP}$$

$$\frac{CB}{CA} = \frac{y}{x} \quad \text{--- (I)}$$

Since $\angle RCB = \angle QBA = 90^\circ$

$\therefore RC \parallel QB$

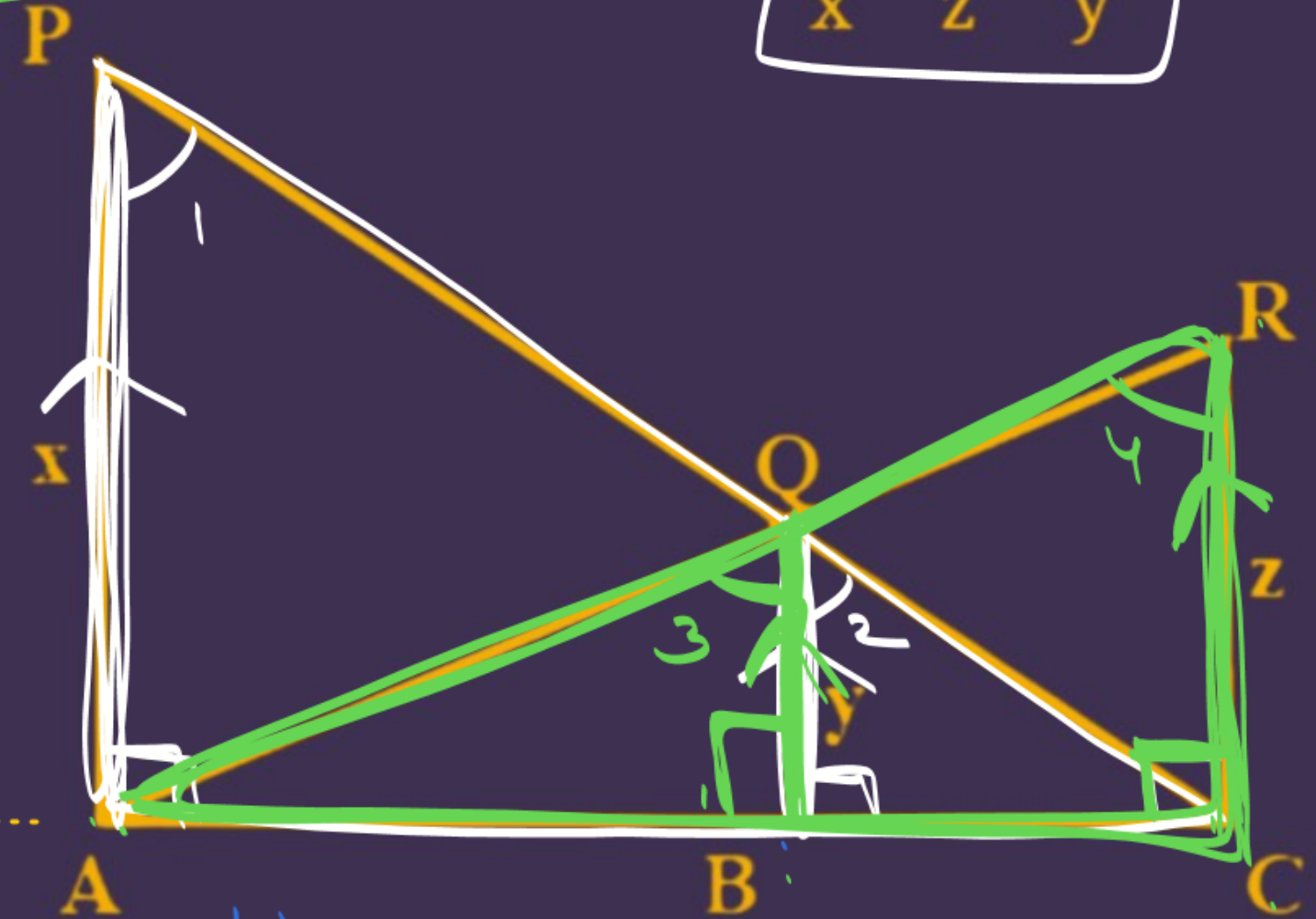
$\therefore \angle RCB = \angle QBA$ (both 90°)

$\angle 3 = \angle 4$ (corr. $\angle$ s)

$\therefore \triangle ABQ \sim \triangle ACR$ (AA)

$$\frac{AB}{AC} = \frac{BQ}{CR}$$

$$\frac{AB}{AC} = \frac{y}{z} \quad \text{--- (II)}$$



(I) + (II)

$$\frac{CB}{CA} + \frac{AB}{AC} = \frac{y}{x} + \frac{y}{z}$$

$$\frac{CB+AB}{AC} = y \left(\frac{1}{x} + \frac{1}{z} \right)$$

$$\frac{AC}{AC} = y \left(\frac{1}{x} + \frac{1}{z} \right)$$

$$\frac{1}{y} = \frac{1}{x} + \frac{1}{z} \quad \text{--- (Proved)}$$

#LP: In a triangle ABC, let P and Q be points on AB and AC respectively such that $PQ \parallel BC$. Prove that the median AD bisects PQ.

Given:- $PQ \parallel BC$, AD $\rightarrow$ median ($BD = DC$)

To prove:- $PR = RQ$

Proof:- $\angle 1 = \angle 2$ (corr.)
 $\angle PAR = \angle BAD$ (common) $\rightarrow \triangle APR \sim \triangle ABD$ (AA)
cpst $\rightarrow \frac{PR}{BD} = \frac{AR}{AD}$ (I)

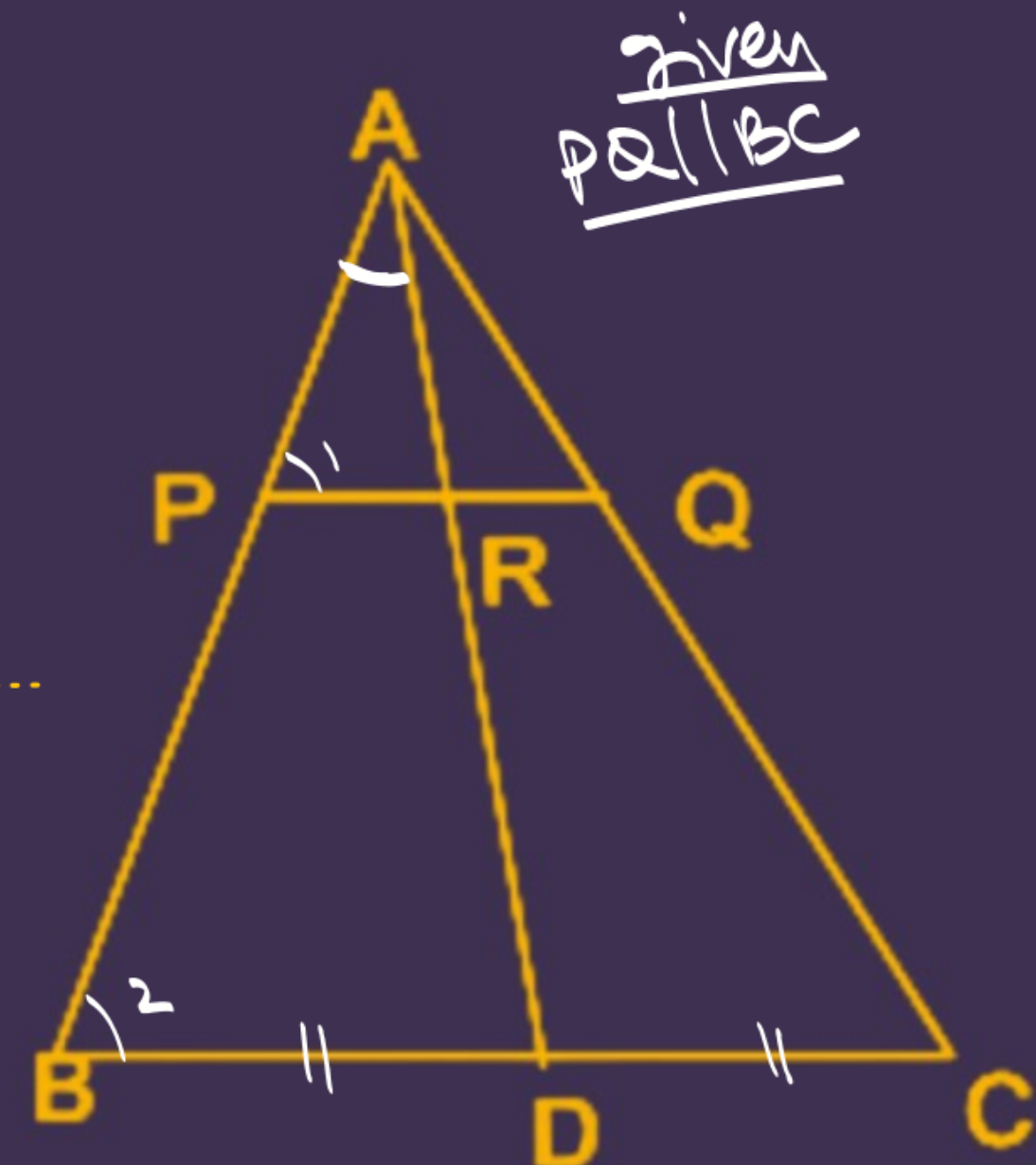
Similarly $\rightarrow \triangle ARQ \sim \triangle ADC$

cpst $\rightarrow \frac{AR}{AD} = \frac{RQ}{DC}$ (II)

from (I) & (II)

$\frac{PR}{BD} = \frac{RQ}{DC}$ ($\because BD = DC$)

$\therefore PR = RQ$ (H/P)



#LP: E is a point on side AD produced of a parallelogram ABCD and BE intersects CD at F. Prove that $\triangle ABE \sim \triangle CFB$.

Given:- ABCD is ||gm.

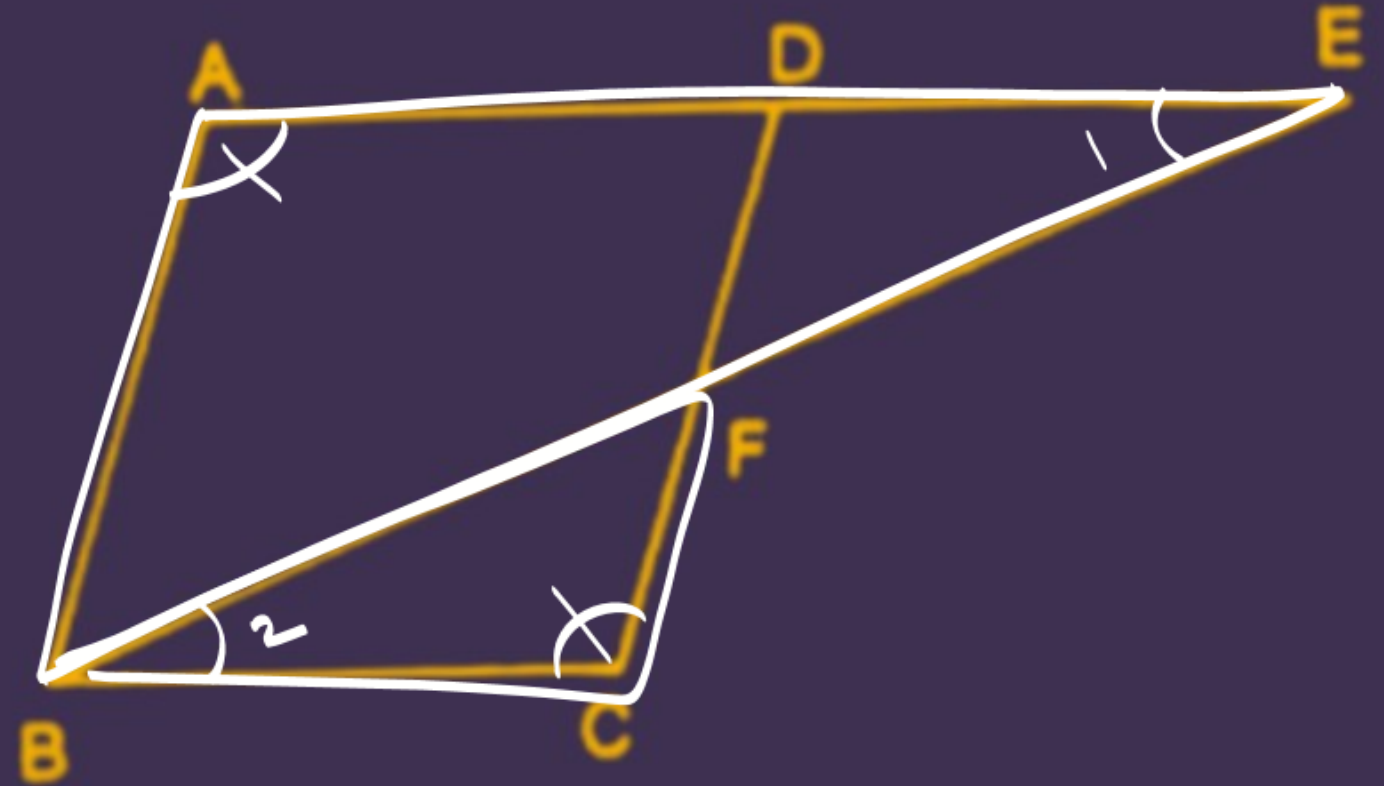
TP $\rightarrow$

Proof:-

$\angle A = \angle C$ (opp $\angle$ s of ||gm equal)

$\angle 1 = \angle 2$ (A Σ A)

AA $\rightarrow$ $\triangle ABE \sim \triangle CFB$ HP



#LP: A right triangle ABC right angled at B. $BD \perp AC$. [CBSE 2009]

To prove:

✓ i. $AB^2 = AD \times AC$

✓ ii. $BC^2 = CD \times AC$

$\triangle CDB \sim \triangle CBA$ (AA)

$\frac{CD}{CB} = \frac{CB}{CA}$

$\Rightarrow CD \cdot CA = CB^2$



(i) $\triangle ADB \sim \triangle ABC$

$\angle A = \angle A$ (common)

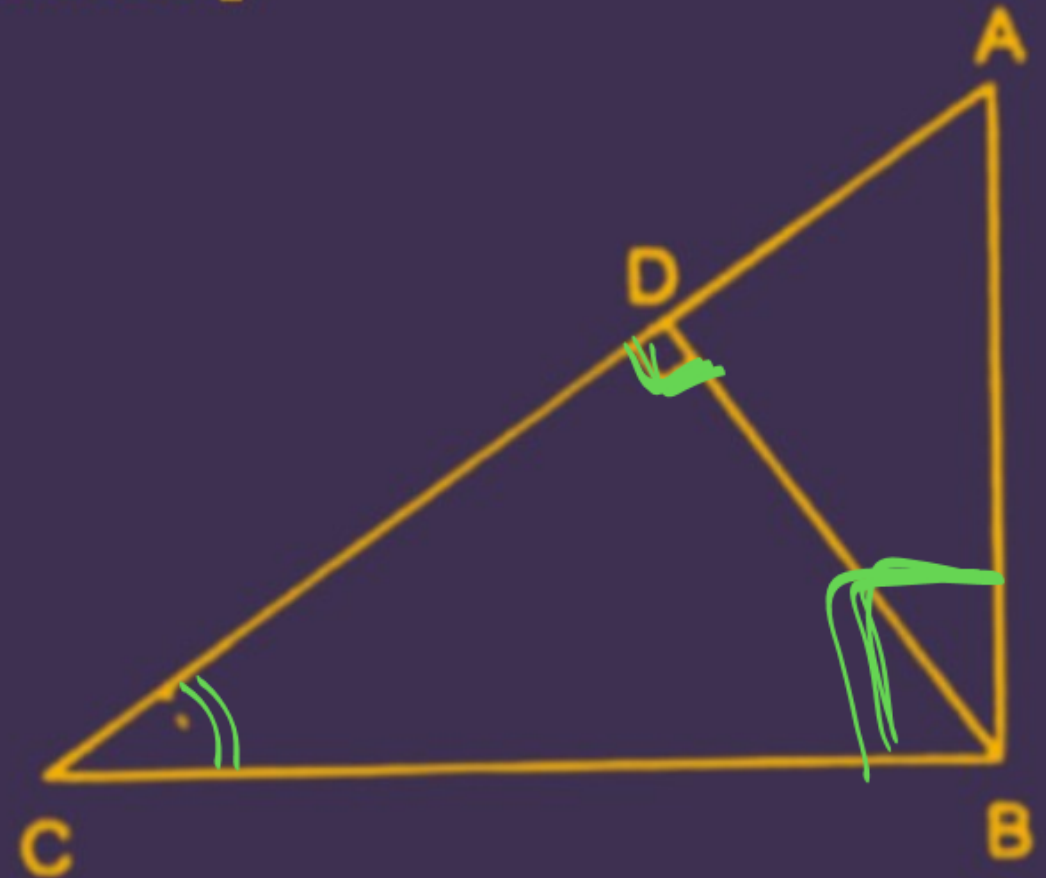
$\angle ADB = \angle ABC$ (both 90°)

AA $\rightarrow \triangle ADB \sim \triangle ABC$

cpst

$\frac{AD}{AB} = \frac{AB}{AC}$

$\Rightarrow AD \cdot AC = AB^2$ (HP)



#LP: If DEFG is a square and $\angle BAC = 90^\circ$. Prove that

(i) $\triangle AGF \sim \triangle DBG$

(ii) $\triangle AGF \sim \triangle EFC$

(iii) $\triangle DBG \sim \triangle EFC$

(iv) $DE^2 = BD \times EC$

$\downarrow$
 $\angle GAF = \angle GDB (90^\circ)$
 $\angle 1 = \angle 2$ (corr. $\angle$)

AA $\triangle AGF \sim \triangle DBG$
 (1P)

$\angle GAF = \angle FEC (90^\circ)$
 $\angle 3 = \angle 4$ (corr. $\angle$)

AA $\rightarrow \triangle AGF \sim \triangle EFC$

cpst
 $\frac{DB}{EF} = \frac{DG}{EC}$

$DB \times EC = DG \times EF$

$DB \times EC = DE \times DE$

$DB \times EC = DE^2$ (1P)

(square) all sides equal



#LP: In Fig, E is a point on side CB produced of an isosceles triangle ABC with $AB = AC$. If $AD \perp BC$ and $EF \perp AC$, prove that

(i) $\triangle ABD \sim \triangle ECF$

(ii) $AB \times EF = AD \times EC$.

[CBSE 2010]

Given:- $\triangle ABC$ is isosc. Δ ($\angle B = \angle C$)
 $AD \perp BC$, $EF \perp AC$

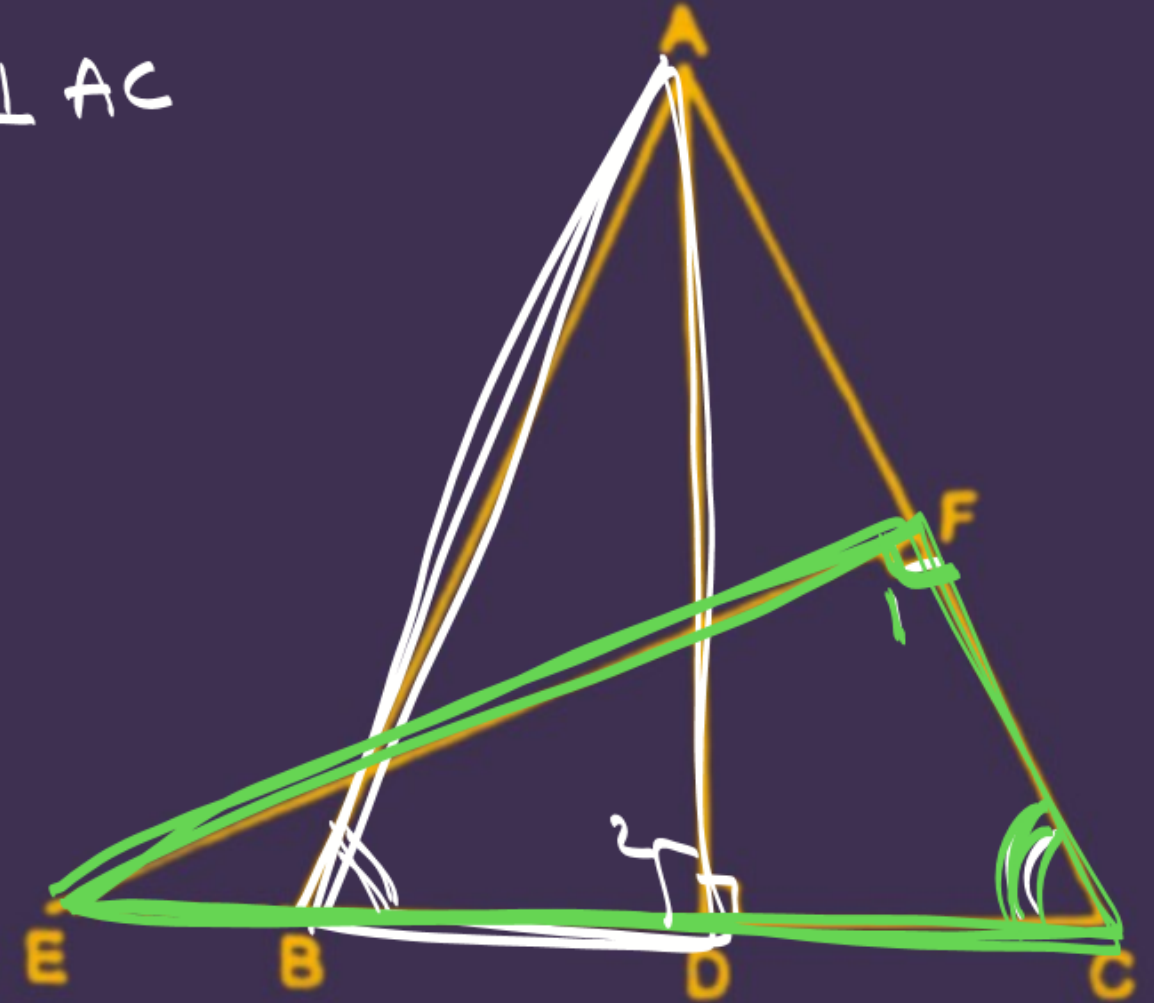
Proof:- (i) $\angle B = \angle C$ (Given)
 $\angle ADB = \angle EFC$ (90°)

$\therefore \triangle ABD \sim \triangle ECF$

∴

$\frac{AB}{EC} = \frac{AD}{EF}$

$AB \times EF = AD \times EC$ (H.P)



#LP: In the given figure, $\triangle FEC \cong \triangle GBD$ and $\angle 1 = \angle 2$, prove that

$\triangle ADE \sim \triangle ABC$.

$$\therefore \angle 1 = \angle 3 \text{ (corr.)}$$

$$\angle 2 = \angle 4 \text{ (corr.)}$$

$$\angle A = \angle A \text{ (common)}$$

$$AAA \Rightarrow \triangle ADE \sim \triangle ABC$$

HP

C.P.C.T

$$EC = BD$$

(I)

(I)

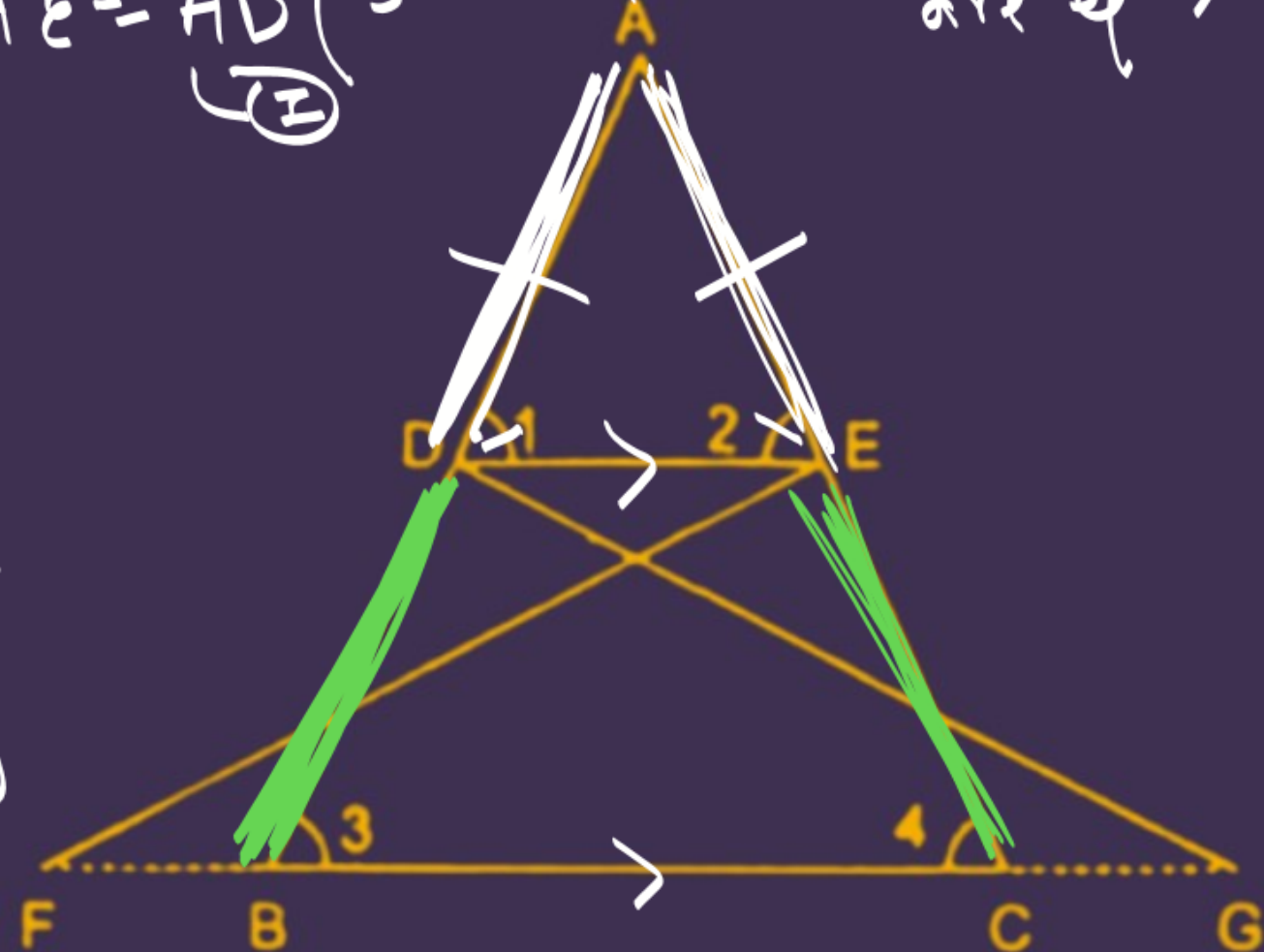
(I)

$$\frac{AE}{EC} = \frac{AD}{BD}$$

$\therefore$ by conv. of BPT

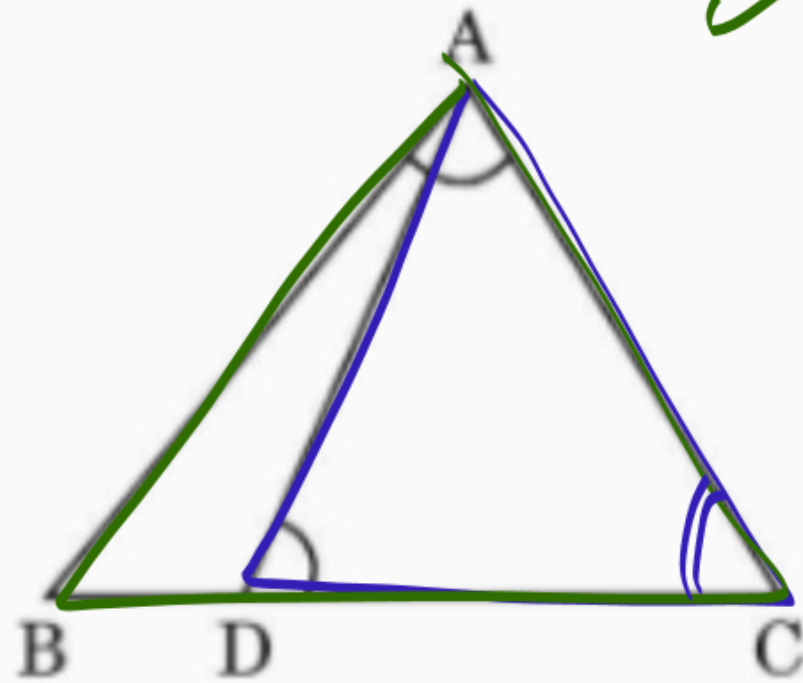
$$DE \parallel BC$$

$AE = AD$ (sides opp to eq. Angles are eq.)
(I)



#LP:

In the given figure, D is a point on the side BC of $\triangle ABC$ such that $\angle ADC = \angle BAC$. Show that $CA^2 = CD \cdot CB$.



$$\angle C = \angle C \text{ (common)}$$

$$\angle ADC = \angle BAC \text{ (given)}$$

$$\therefore \triangle ADC \sim \triangle ABC \text{ (AA)}$$

$$\frac{DC}{AC} = \frac{AC}{BC}$$

$$DC \times BC = AC^2$$

Q.P

[CBSE 2025]

#LP: In the given figure angle $ACB = \text{angle } CDA$, $AC = 8\text{cm}$, $AD = 3\text{cm}$, then BD is

A. $22/3$

B. $26/3$

☒ C. $55/3$

D. $64/3$

$$CA^2 = AD \times AB$$

$$(8)^2 = (3) \times AB$$

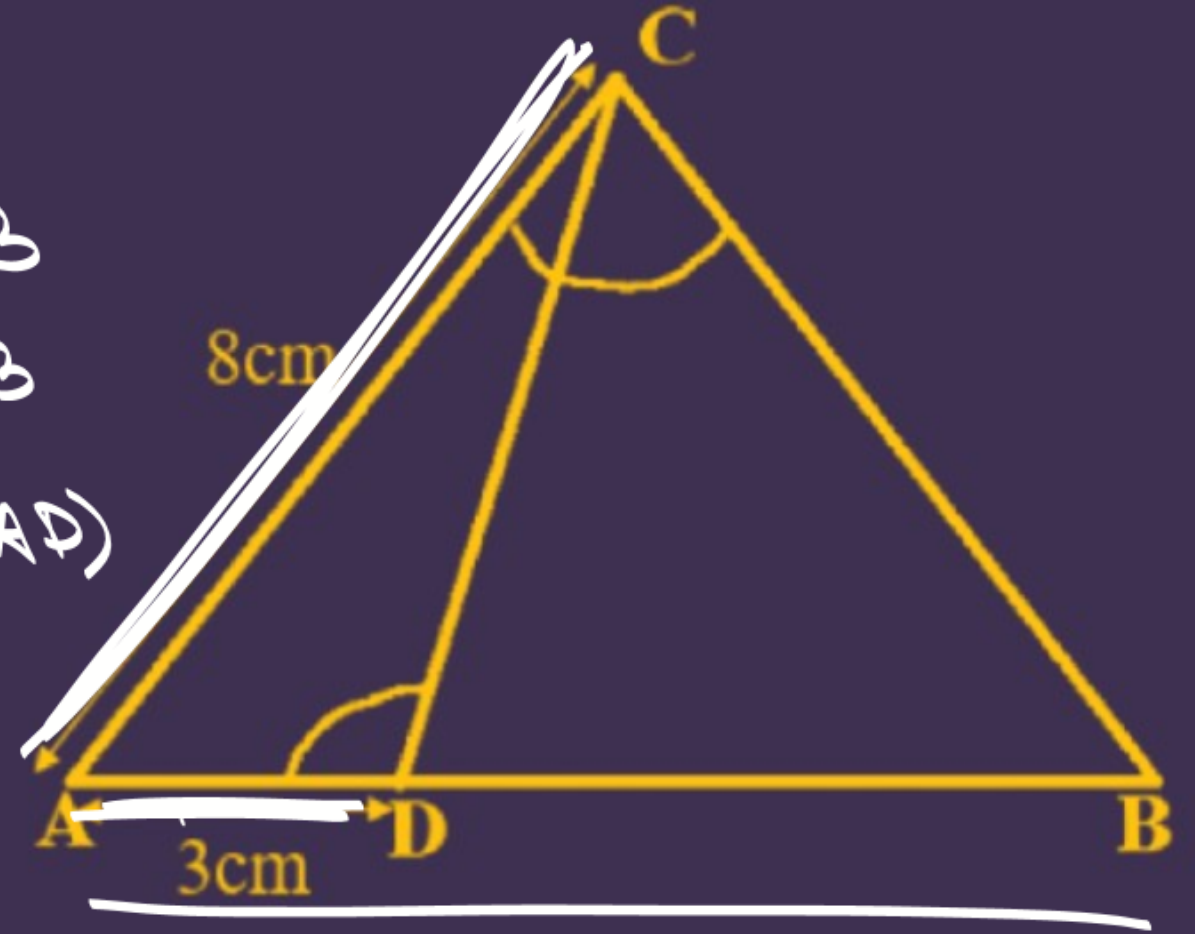
$$64 = 3 \times (BD + AD)$$

$$\frac{64}{3} = BD + 3$$

$$\frac{64}{3} - 3 = BD$$

$$\frac{64 - 9}{3} = BD$$

$$BD = \frac{55}{3}$$



Thank you Goodieess !!

